

SIMATS ENGINEERING

ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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ANNEXURE A COMPARATIVE ANALYSIS

- 1.Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
- 2.Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
- 3.Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
- 4.Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
- 5.Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

*I ANUSHYA DEVI V (192424013) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Comparison of L1, L2, and L3 Cache

Feature	L1 Cache	L2 Cache	L3 Cache
Location	Inside each CPU core	Usually inside/near each core	Shared among CPU cores
Size	Smallest	Medium	Largest
Latency	Very low (fastest)	Low	Higher than L1 and L2
Bandwidth	Highest	High	Lower than L1/L2
Throughput	Highest	High	Moderate
Access Speed	Fastest	Fast	Slower
Typical Size	KBs	Hundreds of KBs to a few MBs	Several MBs to tens of MBs
Main Purpose	Stores frequently used data/instructions	Stores data missed by L1	Reduces accesses to slower main memory

Analysis

- L1 Cache: It has the lowest latency and highest bandwidth, so the CPU can access frequently used instructions and data very quickly. This provides the biggest improvement in processor performance.
- L2 Cache: It is larger than L1 but slightly slower. It stores data that is not found in L1 and helps reduce the number of accesses to RAM.
- L3 Cache: It is the largest cache and is usually shared between CPU cores. Its latency is higher than L1 and L2, but it still provides much faster access than main memory.

Impact on Processor Performance

The cache hierarchy improves CPU performance by keeping frequently needed data close to the processor:

L1 → L2 → L3 → RAM

If data is found in L1, the CPU gets it fastest. If there is an L1 miss, it checks L2, then L3, and finally RAM. Therefore, a high cache hit rate reduces memory waiting time, increases effective throughput, and keeps the CPU busy.

Conclusion

L1 gives the fastest access and highest performance, L2 provides a balance between speed and capacity, while L3 provides large shared storage and reduces expensive RAM accesses.

2. Comparison of SRAM and DRAM

Feature	SRAM	DRAM
Full Form	Static Random Access Memory	Dynamic Random Access Memory
Storage	Uses flip-flops	Uses capacitor and transistor
Latency	Very low (very fast)	Higher than SRAM
Bandwidth	High	Moderate to high
Throughput	High	Lower compared with SRAM
Refresh	Not required	Requires periodic refresh
Cost	Expensive	Cheaper
Density	Low	High
Typical Use	CPU cache (L1, L2, L3)	Main memory (RAM)

Analysis

- **SRAM:** SRAM has very low latency because it does not need refresh operations. Its high bandwidth and throughput make it suitable for CPU cache memory, where the processor needs data very quickly.
- **DRAM:** DRAM has higher latency because it stores data in capacitors and requires periodic refreshing. However, it provides high storage capacity at lower cost, making it suitable for main memory.

Impact on System Performance

SRAM improves processor performance by providing fast access to frequently used data. DRAM provides large memory capacity but is slower, so the processor depends on cache memory to reduce frequent DRAM accesses.

Conclusion

SRAM is faster with lower latency and higher throughput, so it is used for cache memory. DRAM is slower but larger and cheaper, so it is mainly used as the system's main memory.

3. Comparison of Virtual Memory, Paging, and Cache Memory

Feature	Virtual Memory	Paging	Cache Memory
Purpose	Provides more logical memory than physical RAM	Divides virtual memory into fixed-size pages	Stores frequently used data near CPU
Storage	RAM + secondary storage	RAM + secondary storage	SRAM inside/near CPU
Access Latency	Very high when data is on disk	High during page fault	Very low
Bandwidth	Lower	Moderate	Very high
Throughput	Lower when swapping occurs	Reduced during page faults	High
Speed	Slow compared with cache/RAM	Depends on page hit/fault	Fastest
Main Benefit	Supports large programs	Efficient memory management	Speeds up CPU access

Analysis

- **Virtual Memory:** Allows programs to use more memory than the available physical RAM. When required data is not in RAM, it must be obtained from secondary storage, causing very high latency and reducing system throughput.
- **Paging:** Divides memory into fixed-size pages and frames. It allows efficient memory allocation, but a page fault requires data to be loaded from storage, which significantly increases access time.
- **Cache Memory:** Stores frequently accessed instructions and data close to the CPU. It has very low latency and high throughput, allowing the processor to execute instructions faster.

Impact on System Performance

The access-speed relationship is:

Cache → RAM → Virtual Memory/Storage

Cache memory improves throughput by reducing CPU waiting time. Paging and virtual memory improve memory capacity and management, but frequent page faults can reduce system performance.

Conclusion: Cache memory provides the lowest latency and highest throughput, while virtual memory and paging provide better memory capacity and management but can have much higher latency, especially during page faults.

4. Comparison of NAND Flash and DRAM

Feature	NAND Flash	DRAM
Storage Type	Non-volatile	Volatile
Latency	High	Low
Bandwidth	Moderate	High
Throughput	Lower for random access	High
Data Retention	Retains data without power	Loses data when power is removed
Refresh	Not required	Required
Main Use	SSDs, USB drives, storage	Main memory (RAM)
CPU Access	Much slower	Much faster

Analysis

- **NAND Flash:** It is non-volatile, so it can retain data even when the power is off. However, its access latency is much higher than DRAM, especially for random operations. It is mainly used for secondary storage, such as SSDs.
- **DRAM:** It provides much lower latency and higher bandwidth than NAND Flash. It is therefore suitable for main memory, where the CPU requires fast and frequent data access.

Suitability in Hierarchical Memory Systems

A typical memory hierarchy is:

CPU Registers → Cache (SRAM) → DRAM → NAND Flash/SSD

- DRAM is placed closer to the CPU because of its low latency and high throughput.
- NAND Flash is placed farther from the CPU because it has higher latency but provides large, non-volatile, and low-cost storage.

Conclusion

DRAM is better for fast, frequently accessed working data, while NAND Flash is better for large, persistent data storage. Both are important in hierarchical memory systems because they provide a balance between speed, capacity, cost, and data persistence.

5. Comparison of MRAM and RRAM

Feature	MRAM	RRAM
Full Form	Magnetoresistive Random Access Memory	Resistive Random Access Memory
Access Time	Very fast	Fast
Throughput	High	High
Energy Efficiency	High	Very high
Power Consumption	Low, especially in standby	Very low for storage
Volatility	Non-volatile	Non-volatile
Endurance	Very high	High, but technology-dependent
Main Advantage	Fast access and high endurance	Low power and high storage potential

Analysis

- **MRAM:** Uses magnetic states to store data. It provides fast access time, high throughput, and low standby power. It can handle a large number of read/write operations, making it suitable for embedded systems and cache-like applications.
- **RRAM:** Uses changes in electrical resistance to store data. It offers fast access, high density, and excellent energy efficiency. It is especially promising for low-power devices and emerging memory systems.

Performance Comparison

Access Time: MRAM generally provides very fast access, while RRAM can also provide fast access but varies depending on the technology.

Throughput: Both technologies can provide high throughput. MRAM is particularly suitable for applications requiring frequent read/write operations.

Energy Efficiency: RRAM can achieve very low energy consumption, while MRAM also offers good energy efficiency because it does not require continuous refresh.

Conclusion

MRAM is preferred for fast access and high endurance, whereas RRAM is attractive for low-power and high-density applications. Both are non-volatile memories that can reduce dependence on traditional DRAM and NAND Flash in future hierarchical memory systems.